

Операционные системы

Командная оболочка Midnight Commander

Малютина Софья Александровна

2026-09-01

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

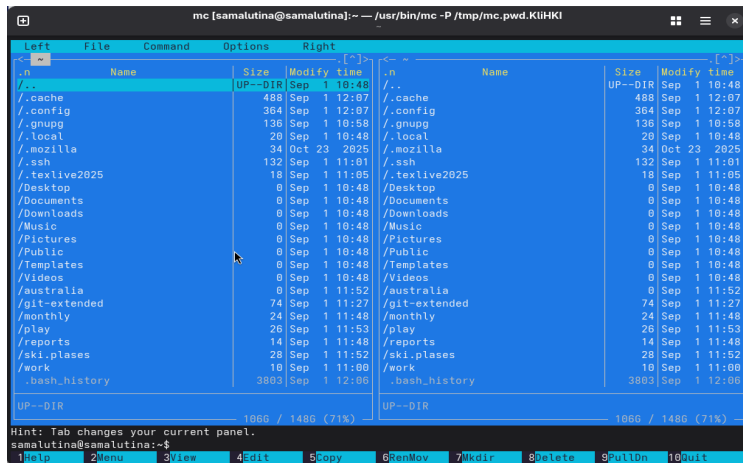


Рисунок 1: Запуск mc

Работа с Midnight Commander

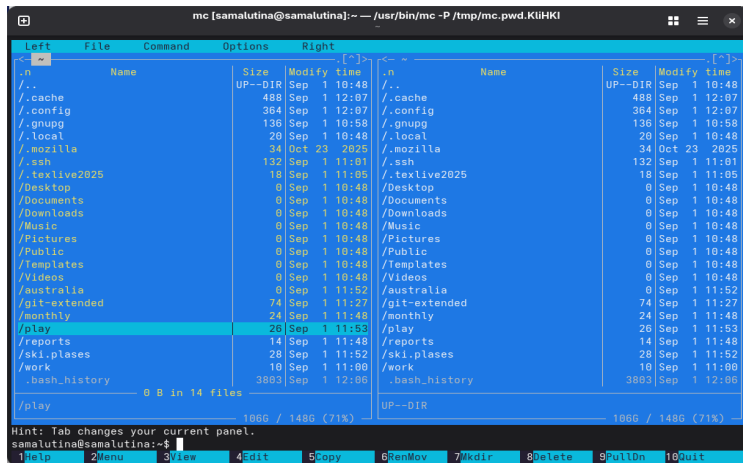


Рисунок 2: Выделение

Работа с Midnight Commander

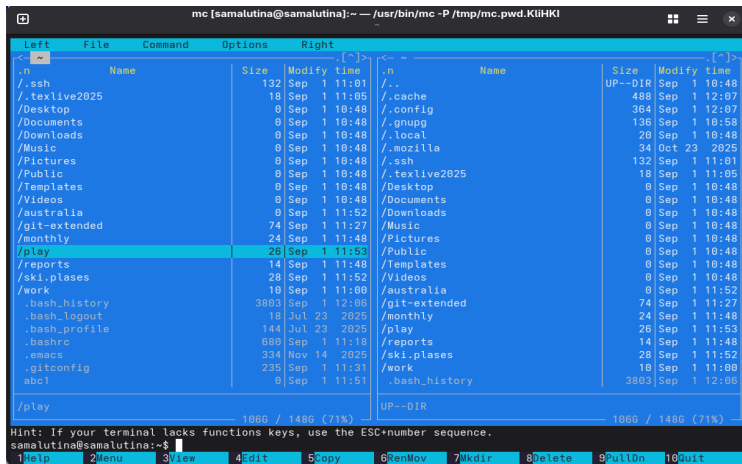


Рисунок 3: Отмена

Работа с Midnight Commander

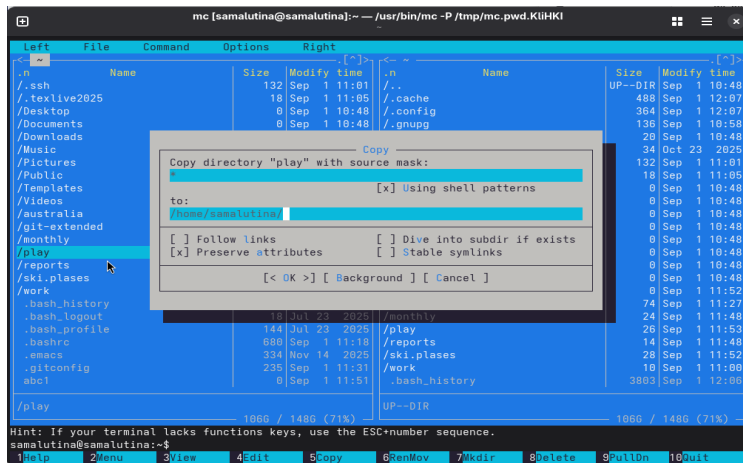


Рисунок 4: Копирование

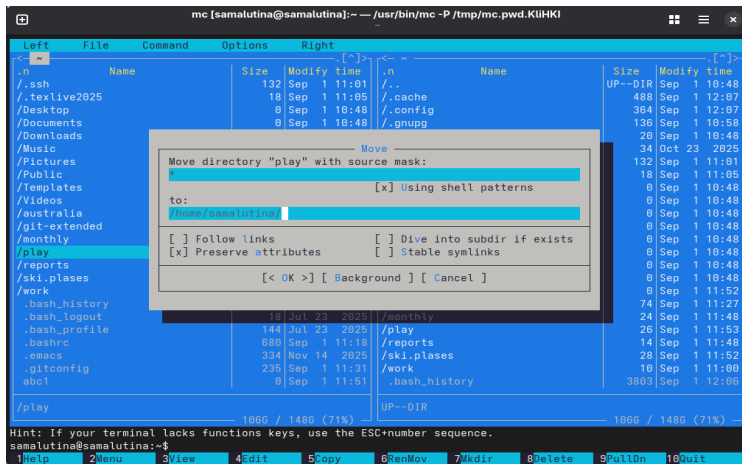


Рисунок 5: Перемещение

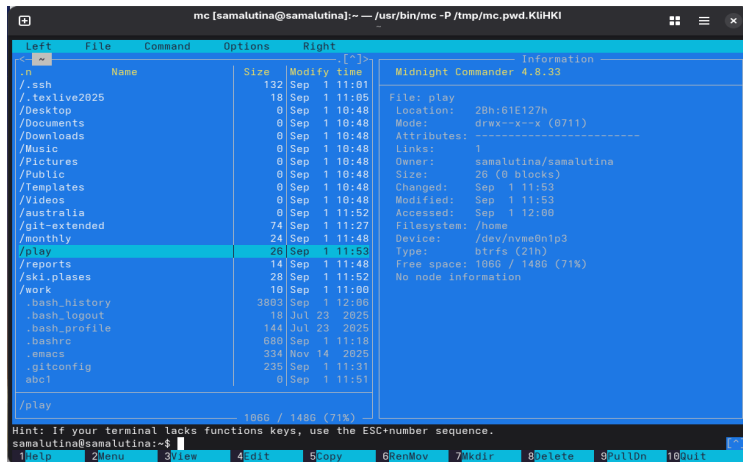


Рисунок 6: Информация

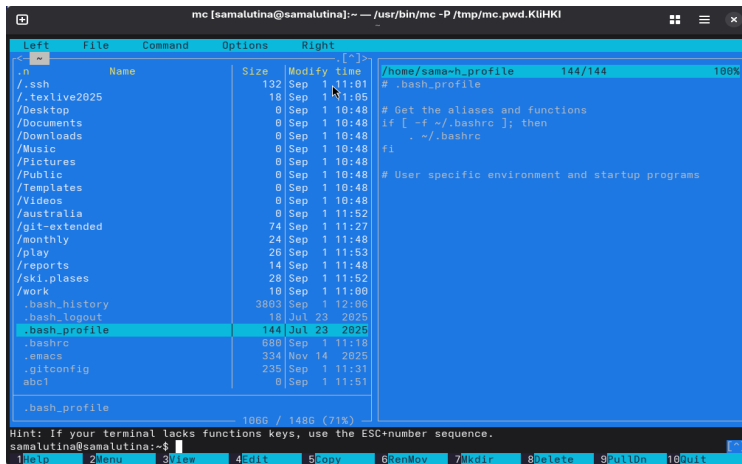


Рисунок 7: Быстрый просмотр

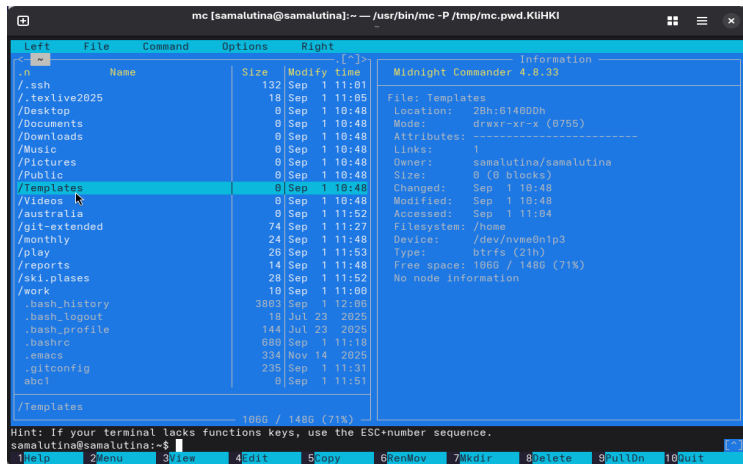


Рисунок 8: Информация

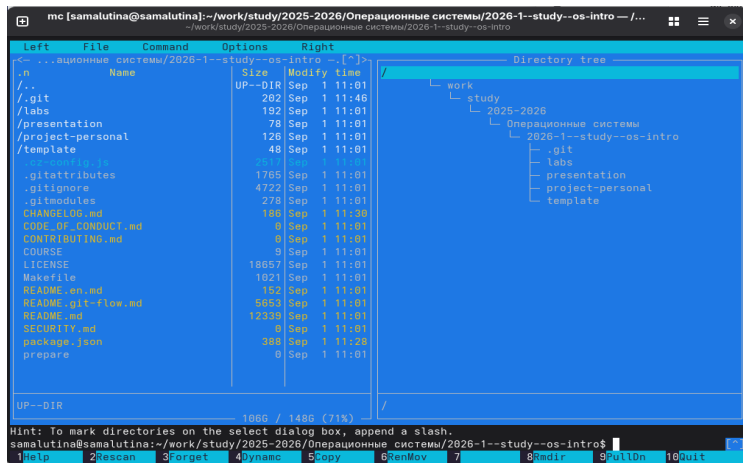
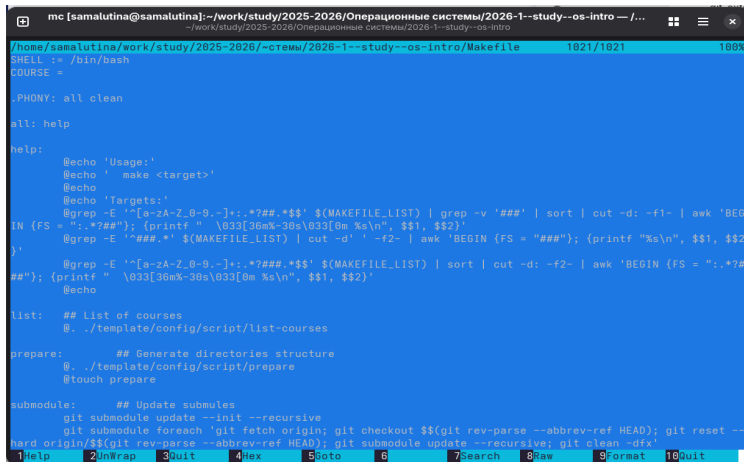


Рисунок 9: Дерево каталогов



```
mc [samalutina@samalutina]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
/home/samalutina/work/study/2025-2026/~стемы/2026-1--study--os-intro/Makefile 1021/1021 100%
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

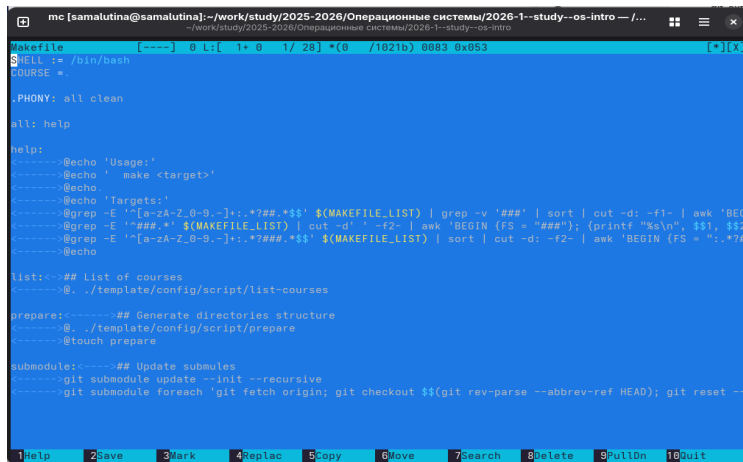
help:
    @echo 'Usage:'
    @echo ' make <target>'
    @echo
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9.~]+:.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ":.*?";} {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9.~]+:.*$$' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###";} {printf "%s\n", $$1, $$2}'
    @echo
list:    ## List of courses
    @. ./template/config/script/list-courses

prepare:    ## Generate directories structure
    @. ./template/config/script/prepare
    @touch prepare

submodule:    ## Update submodules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-parse --abbrev-ref HEAD); git submodule update --recursive; git clean -dfx'
```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander



```
mc [samalutina@samalutina]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro

Makefile      [----] 0 L:[ 1+ 0 1/ 28] *(0 /1021b) 0083 0x053 [*)(X]
SHELL := /bin/bash
COURSE *.

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo.
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN
----->@grep -E '^[^##.*] $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN (FS = "###"); {printf "%s\n", $$1, $$2
----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN (FS = ".*?#
----->@echo

list:<-->## List of courses
----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<---->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

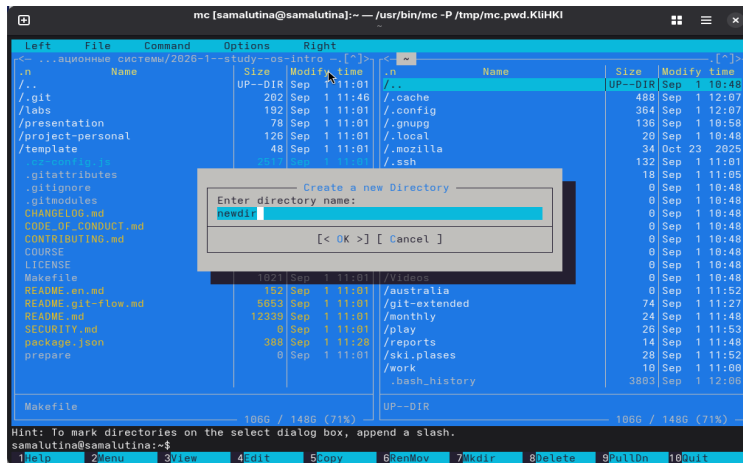


Рисунок 12: Создание каталога

Работа с Midnight Commander

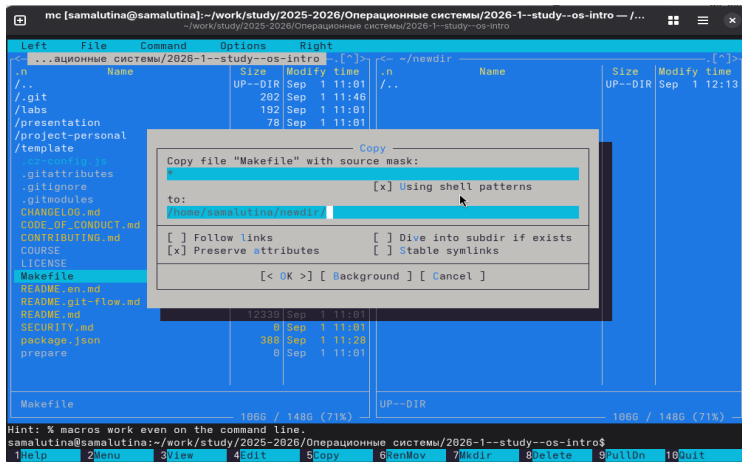


Рисунок 13: Копирование в файлов в созданный каталог

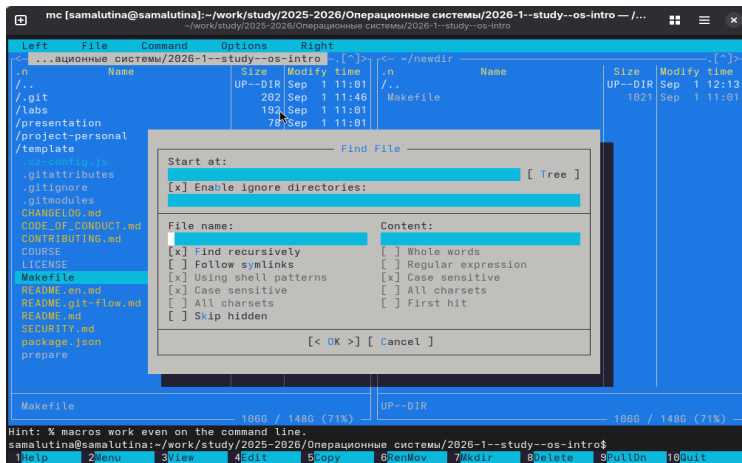


Рисунок 14: Поиск файлов

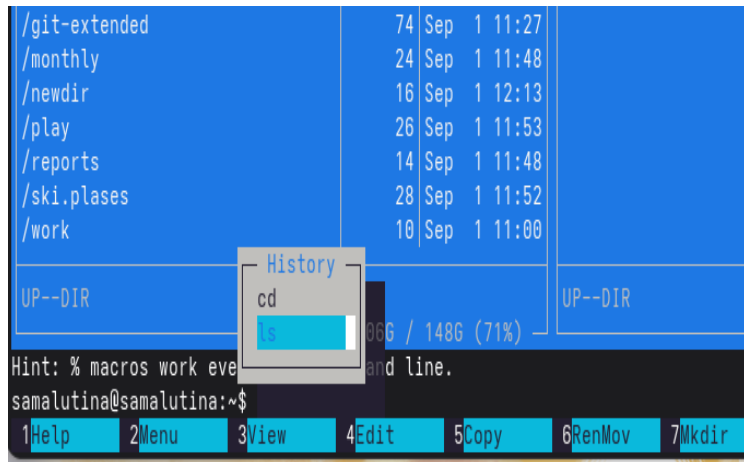


Рисунок 15: История команд

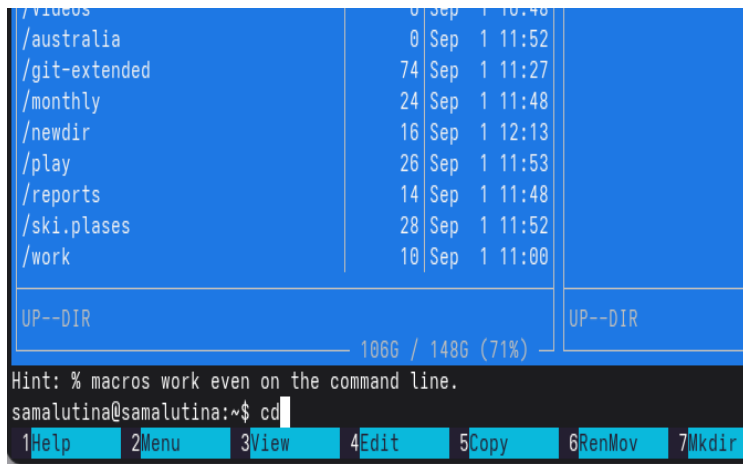
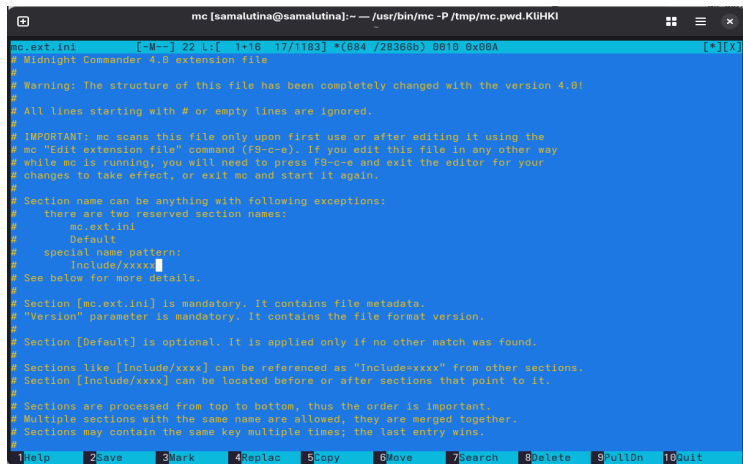


Рисунок 16: Переход в домашний каталог



```
mc [samalutina@samalutina]:~ — /usr/bin/mc -P /tmp/mc.pwd.KUHKl
mc.ext.ini  [-M--] 22 L:[ 1+16 17/1183] *(684 /28366b) 0010 0x00A [*][X]
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 17: Просмотр файла расширений

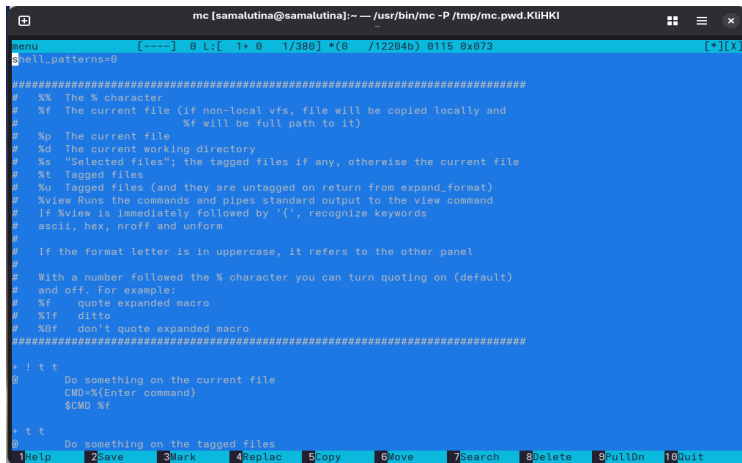


Рисунок 18: Просмотр файла меню

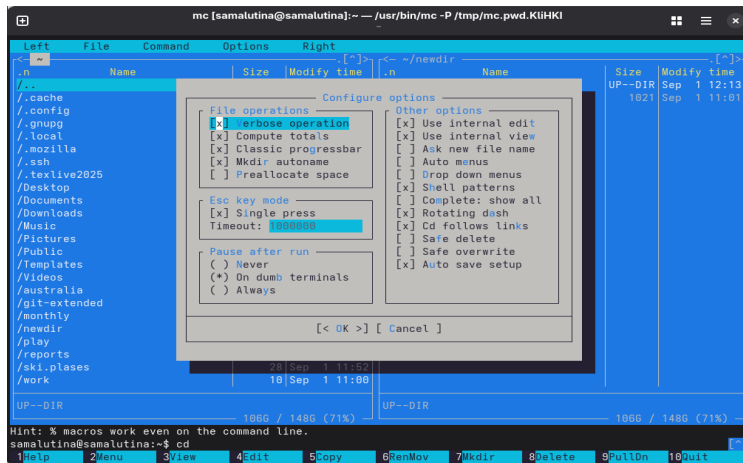


Рисунок 19: Конфигурация

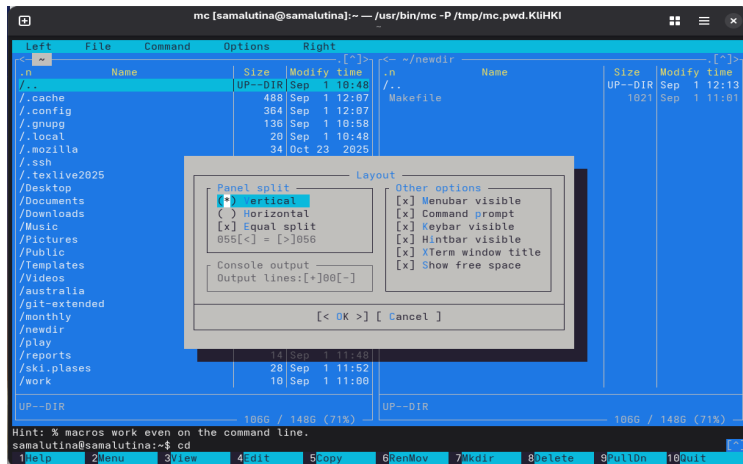


Рисунок 20: Внешний вид

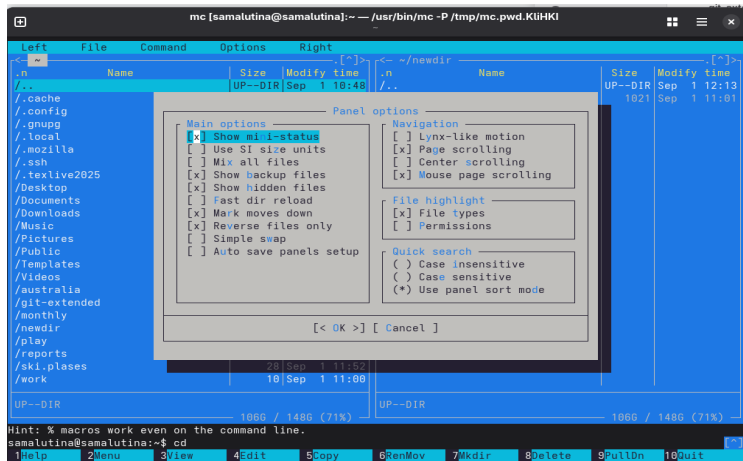


Рисунок 21: Настройки панелей

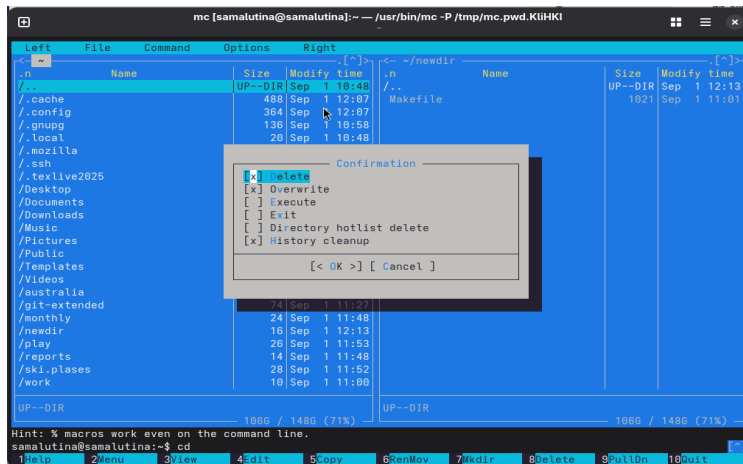


Рисунок 22: Подтверждение

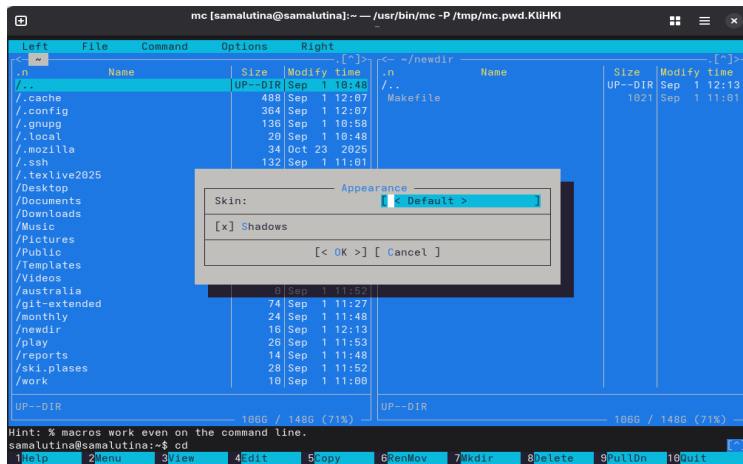


Рисунок 23: Оформление

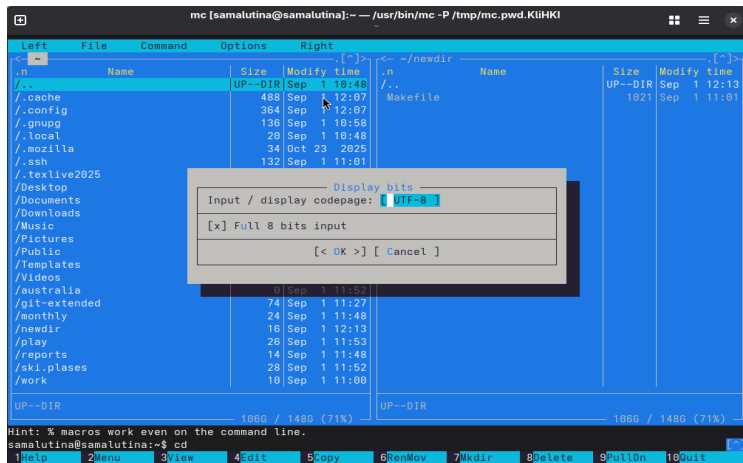


Рисунок 24: Кодировка символов

Работа с Midnight Commander

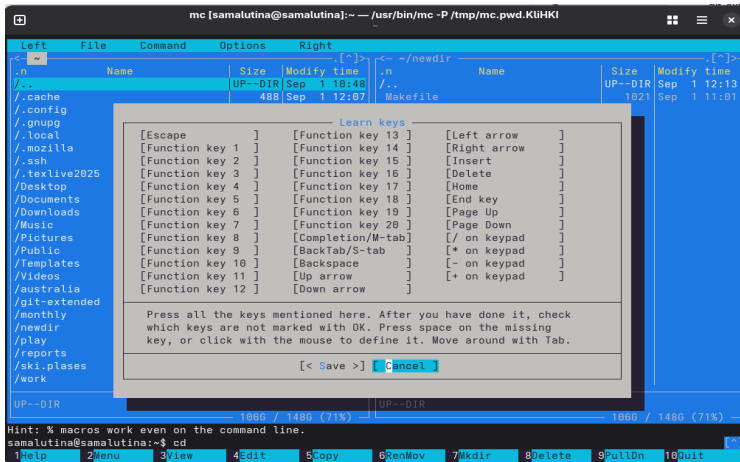
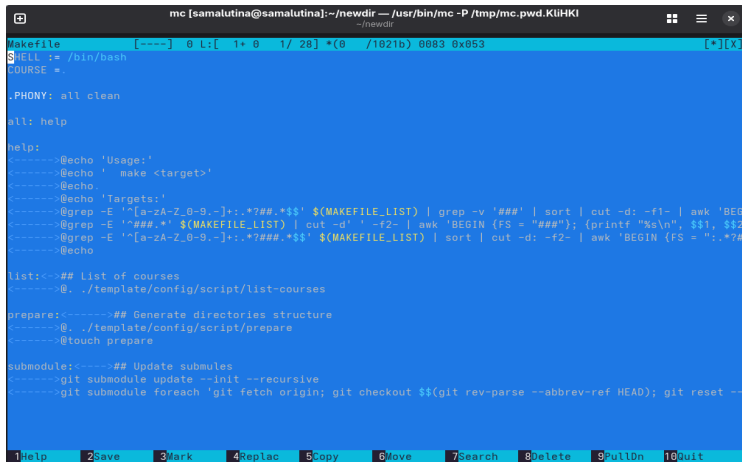


Рисунок 25: Распознавание клавиш

Работа с редактором Midnight Commander



```
mc [samalutina@samalutina]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.KUHKI
~newdir

Makefile      [----]  0 L: [  1+ 0  1/ 28]  *(0  /1021b) 0083 0x053  [*][x]

SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN
<----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d: ' ' -f2- | awk 'BEGIN {FS = "###"; {printf "%s\n", $$1, $$2
<----->@grep -E '^[a-zA-Z_0-9-]+.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##
<----->@echo

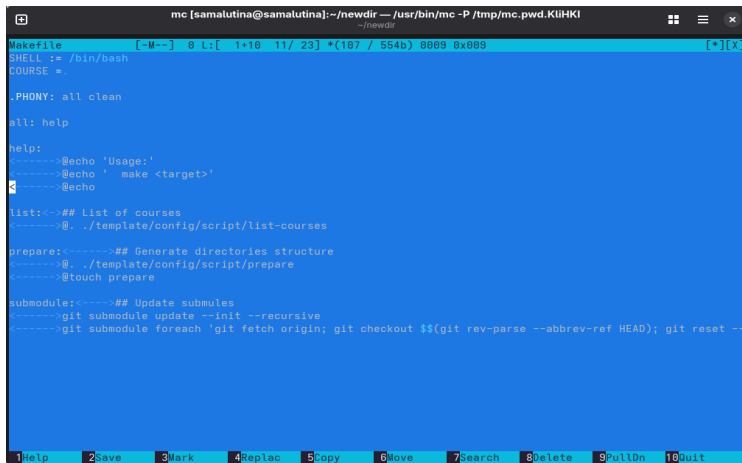
list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<---->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --
```

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander



```
mc [samalutina@samalutina]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.KUHKI
~/newdir
Makefile      [-M--]  0 L:[ 1+10 11/ 23] *(107 / 554b) 0000 0x000 [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo

list:<-->## List of courses
----->@. ./template/config/script/list-courses

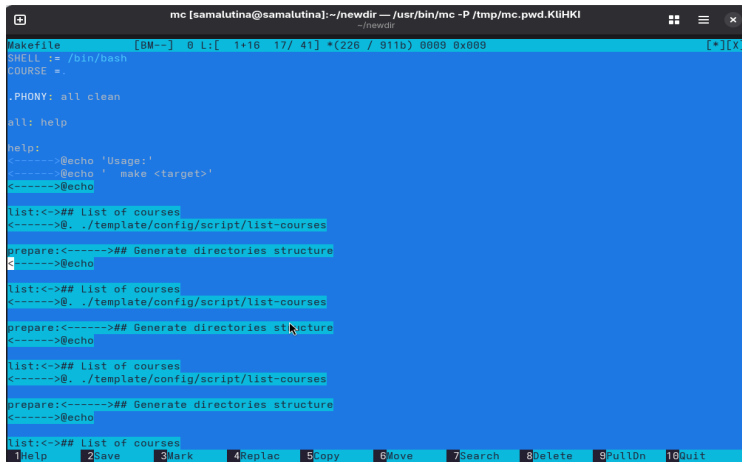
prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<---->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --
```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander interface. The title bar indicates the user is 'samalutina@samalutina' in the directory '~/.newdir'. The main window displays a file named 'Makefile' with a status bar at the top showing '[BM--] 0 L: [1+16 17/ 41] *(226 / 911b) 8000 0x000'. The file content is a Makefile script. The menu bar at the bottom contains 10 items: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, and 10Quit.

```
mc [samalutina@samalutina]: ~/.newdir — /usr/bin/mc -P /tmp/mc.pwd.KüHKl
~/.newdir

Makefile [BM--] 0 L: [ 1+16 17/ 41] *(226 / 911b) 8000 0x000 [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 28: Копирование фрагмента

Работа с редактором Midnight Commander

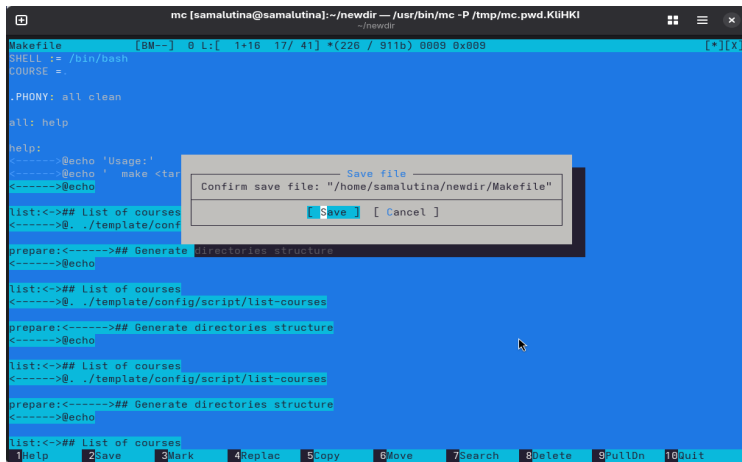
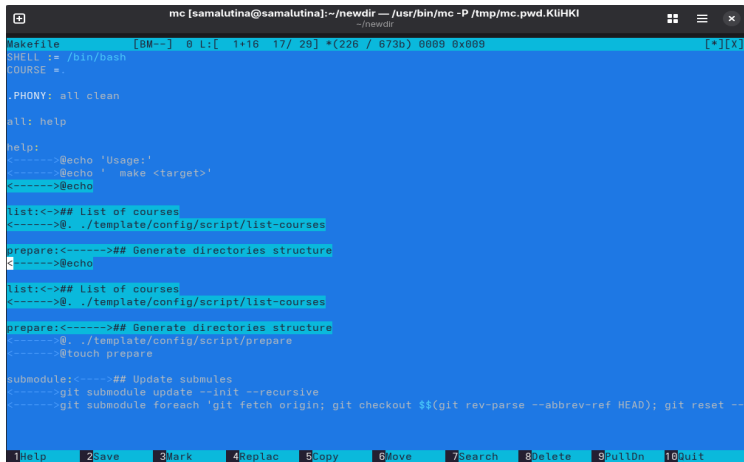


Рисунок 29: Сохранение

Работа с редактором Midnight Commander



```
mc [samalutina@samalutina]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.KUHKI
~newdir

Makefile [BM--] 0 L: [ 1+16 17/ 29] *(226 / 673b) 0009 0x009 [*)(x]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

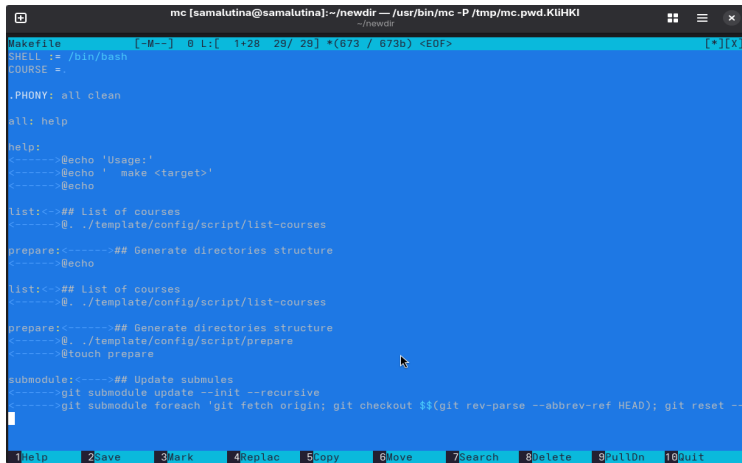
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<---->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --
```

1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit

Рисунок 30: Отмена

Работа с редактором Midnight Commander



```
mc [samalutina@samalutina]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.KtHkI
~/newdir
Makefile  [-M--]  0 L:[ 1+28 29/ 29] *(673 / 673b) <EOF>  [*](x)
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

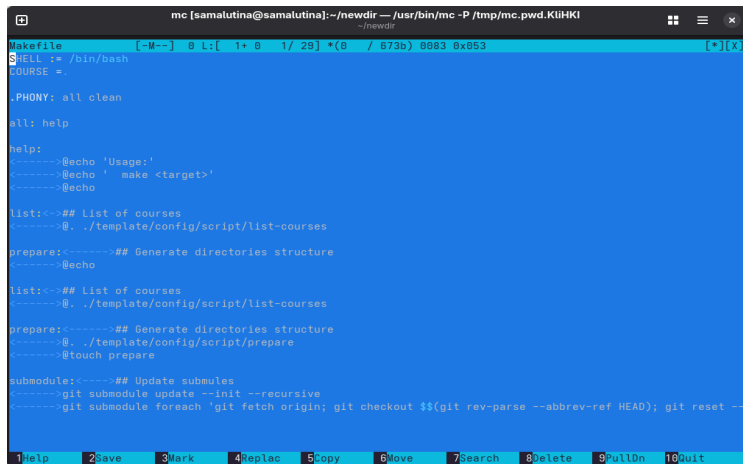
list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<---->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --
```

1help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 31: Переход в конец файла



```
mc [samalutina@samalutina]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.KlHkI
~/newdir
Makefile      [-M--]  8 L:[ 1+ 8  1/ 29]  *(8  / 673b)  0083  0x053  [*][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@echo

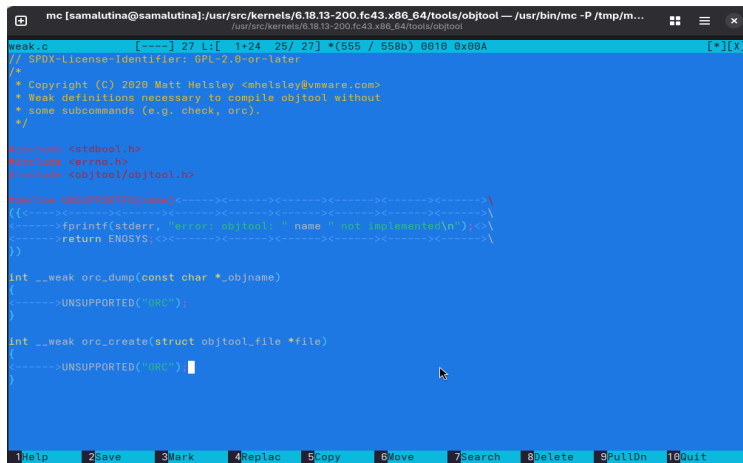
list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --
```

Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander



```
mc [samalutina@samalutina]:/usr/src/kernels/6.18.13-200.fc43.x86_64/tools/objtool — /usr/bin/mc -P /tmp/m...
/usr/src/kernels/6.18.13-200.fc43.x86_64/tools/objtool

weak.c      [----] 27 L:[ 1+24 25/ 27] *(555 / 558b) 0010 0x00A [*][X]
// SPDX-License-Identifier: GPL-2.0-or-later
/*
 * Copyright (C) 2020 Matt Helsley <mhelsley@vmware.com>
 * Weak definitions necessary to compile objtool without
 * some subcommands (e.g. check, orc).
 */

#include <stdbool.h>
#include <errno.h>
#include <objtool/objtool.h>

#define UNSUPPORTED(name) <-----><-----><-----><-----><-----><----->\
((<-----><-----><-----><-----><-----><-----><-----><----->\
<----->fprintf(stderr, "error: objtool: " name " not implemented\n");<-\
<----->return ENOSYS;<-----><-----><-----><-----><-----><----->\
))

int __weak orc_dump(const char *_objname)
{
    <----->UNSUPPORTED("orc");
}

int __weak orc_create(struct objtool_file *file)
{
    <----->UNSUPPORTED("orc");
}
```

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

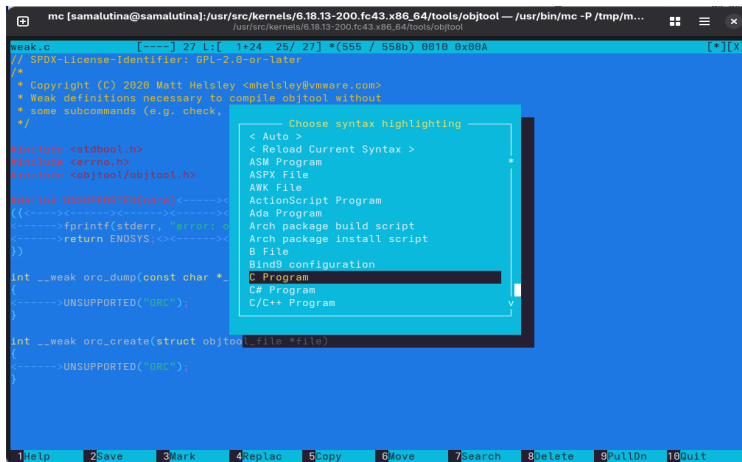


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.